

# Intranet Asset Discovery and Server Security Hardening Based on a Virtualized Environment

## 1. Setting Up an Isolated Lab Environment

Install:

Oracle VirtualBox

DownloadL:

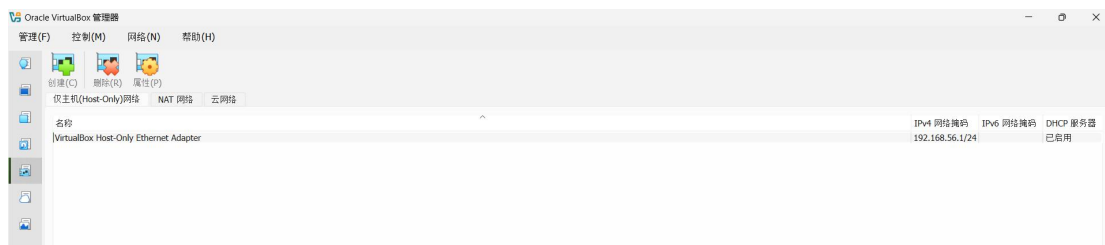
kali-linux-2026.2-installer-amd64.iso

ubuntu-22.04.5-live-server-amd64.iso

## 2. Network Planning

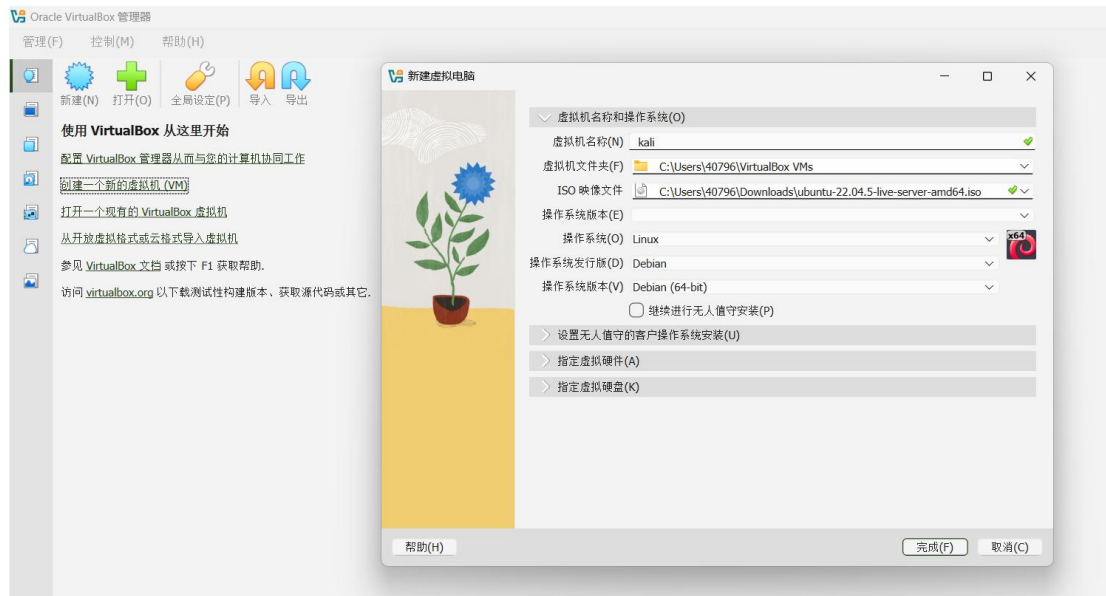
Since this experiment does not involve the external network, Host-only mode is used.

Create an isolated internal network (DHCP can be enabled or disabled).



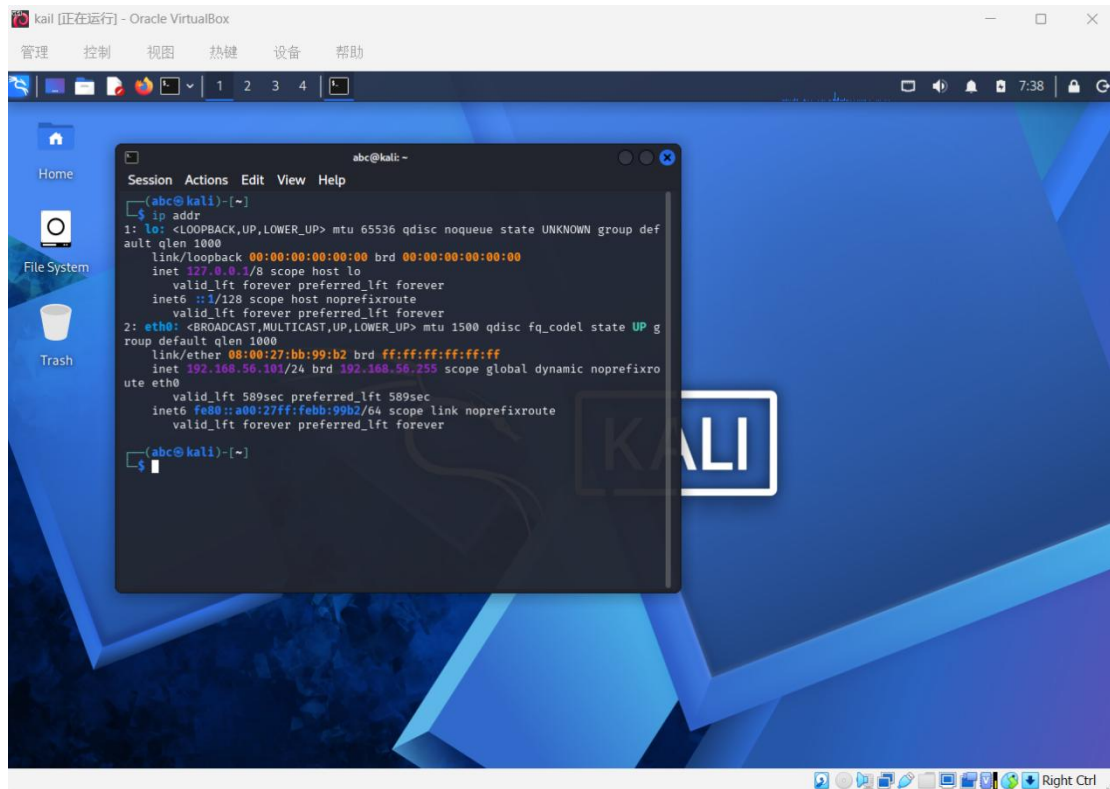
Create two virtual machines: Kali and Ubuntu. The memory for both is set to 2048MB (Note: corrected from '2048KB' to

'2048MB' for system viability), CPU is 2 vCPUs, virtual hard disk is 20GB, and the disk file type and format are VDI.



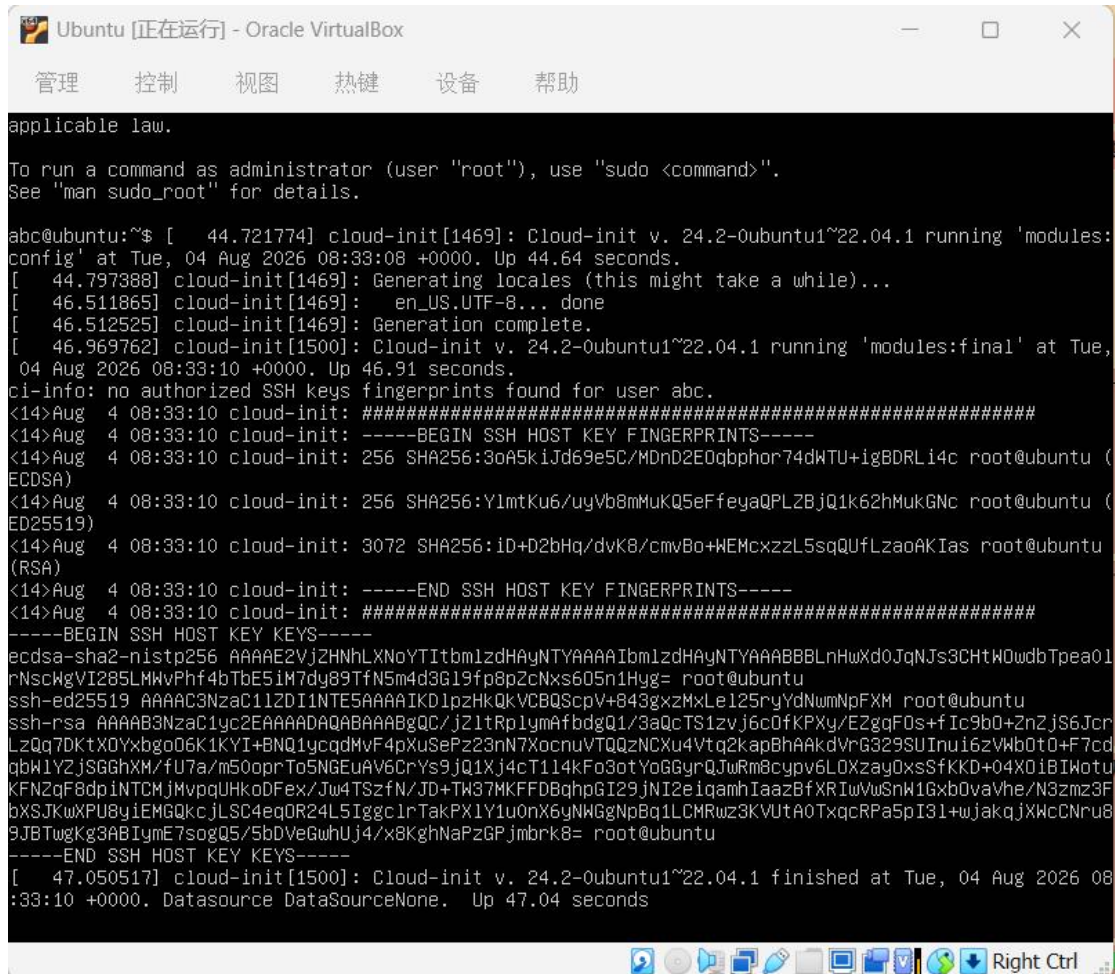
Start Kali using the Graphical User Interface (GUI) installer. During the installation, select the appropriate language, region, etc. Leave the domain name blank, and select the beginner-friendly "Guided - use entire disk" for disk partitioning.

After installation, open the terminal and input ip addr to verify the IP address.



When installing Ubuntu, remember to check the option to install the OpenSSH server. The process is relatively straightforward and will not be detailed here.

Upon successful first login, the background program automatically prints the logs:



```
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

abc@ubuntu:~$ [ 44.721774] cloud-init[1469]: Cloud-init v. 24.2-0ubuntu1~22.04.1 running 'modules:
config' at Tue, 04 Aug 2026 08:33:08 +0000. Up 44.64 seconds.
[ 44.797388] cloud-init[1469]: Generating locales (this might take a while)...
[ 46.511865] cloud-init[1469]: en_US.UTF-8... done
[ 46.512525] cloud-init[1469]: Generation complete.
[ 46.969762] cloud-init[1500]: Cloud-init v. 24.2-0ubuntu1~22.04.1 running 'modules:final' at Tue,
04 Aug 2026 08:33:10 +0000. Up 46.91 seconds.
ci-info: no authorized SSH keys fingerprints found for user abc.
<14>Aug 4 08:33:10 cloud-init: #####
<14>Aug 4 08:33:10 cloud-init: -----BEGIN SSH HOST KEY FINGERPRINTS-----
<14>Aug 4 08:33:10 cloud-init: 256 SHA256:3oA5kiJd69e5C/MDnD2E0qbphor74dWTU+igBDRLi4c root@ubuntu (
ECDSA)
<14>Aug 4 08:33:10 cloud-init: 256 SHA256:Y1mtKu6/uyVb8mMuKQ5eFfeyaQPLZBJQ1k62hMukGNc root@ubuntu (
ED25519)
<14>Aug 4 08:33:10 cloud-init: 3072 SHA256:iD+D2bHq/dvK8/cmvBo+WEMcxzzL5sqQUfLzaoAKIas root@ubuntu
(RSA)
<14>Aug 4 08:33:10 cloud-init: -----END SSH HOST KEY FINGERPRINTS-----
<14>Aug 4 08:33:10 cloud-init: #####
-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBLnHwXd0JqNJs3CHtWQwdbTpea01
rNscWgVI285LMWvPhf4bTbE5iM7dy89TfN5m4d3G19fp8p2cNxs605n1Hyg= root@ubuntu
ssh-ed25519 AAAAC3NzaC1l2DI1NTE5AAAAIKDlpzHkKqVCBQScpV+843gxzMxLe125ryYdNmNpFXM root@ubuntu
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGC/jZ1tRplymAfbdgQ1/3aQcTS1zvJ6c0fKPxY/EZggF0s+fIc9b0+ZnZjS6Jcr
LZQg7DKtX0YxbgoD6K1KYI+BNQ1ycqdMvF4pXuSePz23nN7XocnuVTQQzNCXu4Vtq2kapBhAAkdVrG329SUIInui6zVWb0t0+F7cd
qbW1Y2JSGGhXM/fU7a/m50oprTo5NGEuAV6CrYs9Jq1Xj4cT114kFo3otYoGGYrQJwRm8cypv6L0XzayDxsSfKKD+04X0iBIWotu
KFNZqF8dpINTCMjMvpqUHkoDFex/Jw4TSzfN/JD+TW37MKFFDBqhpGI29jNI2eiqamhIaazBfXRiWwSnW1Gxb0vaVhe/N3zmz3F
bXSJKwXPU8yiEMGQkcjLSC4eqQR24L5IggclrTakPX1Y1u0nX6yNWGgNpBq1LCMRwz3KVUtA0TxqcRPa5pI3l+wjakqjXWcCNru8
9JBtwgKg3ABImE7sogQ5/5bDVeGwhUj4/x8KghNaPzGPjmbk8= root@ubuntu
-----END SSH HOST KEY KEYS-----
[ 47.050517] cloud-init[1500]: Cloud-init v. 24.2-0ubuntu1~22.04.1 finished at Tue, 04 Aug 2026 08
:33:10 +0000. Datasource DataSourceNone. Up 47.04 seconds
```

### 3. Configuring Target Machine Services

Check the IP address (e.g., 192.168.56.102/24) and verify the SSH status.

```

abc@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d5:8b:1f brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s3
        valid_lft 409sec preferred_lft 409sec
    inet6 fe80::a00:27ff:fed5:8b1f/64 scope link
        valid_lft forever preferred_lft forever
abc@ubuntu:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2026-08-04 08:32:33 UTC; 6h ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 854 (sshd)
     Tasks: 1 (limit: 2219)
    Memory: 2.9M
       CPU: 17ms
    CGroup: /system.slice/ssh.service
            └─854 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 04 08:32:33 ubuntu systemd[1]: Starting OpenBSD Secure Shell server...
Aug 04 08:32:33 ubuntu sshd[854]: Server listening on 0.0.0.0 port 22.
Aug 04 08:32:33 ubuntu sshd[854]: Server listening on :: port 22.
Aug 04 08:32:33 ubuntu systemd[1]: Started OpenBSD Secure Shell server.
abc@ubuntu:~$

```

After that, temporarily switch the Ubuntu network adapter to NAT mode. Once the network refreshes, update the software sources in the terminal sequentially:

```
sudo apt update
```

One-click install Apache Web Server and MySQL Database:

```
sudo apt install -y apache2 mysql-server
```

Start the Apache service:

```
sudo systemctl start apache2
```

Once completed, check the active ports (Note: netstat -tunlp is a legacy command and requires pre-installation to work;

alternatively, you can use `sudo ss -tunlp` to view active port details).

```
abc@ubuntu:~$ netstat -tunlp
Command 'netstat' not found, but can be installed with:
sudo apt install net-tools
abc@ubuntu:~$ sudo ss -tunlp
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port
Process
udp        UNCONN     0           0           127.0.0.53%lo:53        0.0.0.0:*
users:((("systemd-resolve",pid=1823,fd=13))
udp        UNCONN     0           0           10.0.2.15%enp0s3:68     0.0.0.0:*
users:((("systemd-network",pid=4388,fd=18))
tcp        LISTEN     0           4096        127.0.0.53%lo:53        0.0.0.0:*
users:((("systemd-resolve",pid=1823,fd=14))
tcp        LISTEN     0           128        0.0.0.0:22              0.0.0.0:*
users:((("sshd",pid=854,fd=3))
tcp        LISTEN     0           70         127.0.0.1:33060         0.0.0.0:*
users:((("mysqld",pid=3307,fd=21))
tcp        LISTEN     0           151        127.0.0.1:3306         0.0.0.0:*
users:((("mysqld",pid=3307,fd=23))
tcp        LISTEN     0           511        *:80                    *:80
users:((("apache2",pid=3714,fd=4),("apache2",pid=3713,fd=4),("apache2",pid=3712,fd=4))
tcp        LISTEN     0           128        [::]:22                 [::]:*
users:((("sshd",pid=854,fd=4))
abc@ubuntu:~$
```

As shown in the output, port 22 (SSH service, allowing connections from any network interface), port 80 (Web service, used by Apache to serve web pages), and port 3306 (MySQL database, allowing other programs to connect and manipulate the database via the network) are all working normally. (Note: For security reasons, Ubuntu 22.04 sets the default bind-address of MySQL to 127.0.0.1, which restricts database access to the local machine only.)

#### 4.Asset Discovery and Security Scanning

Switch the Ubuntu network adapter back to Host-only mode. Switch back to the Kali virtual machine, open the terminal, and prepare to perform scanning from the attacker machine.

The role of Nmap is to scan the entire network to discover open ports and check for vulnerabilities. The more ports that are open, the more potential entry points are available for attackers to exploit.

Enter the following in the Kali terminal to perform host discovery:

```
nmap -sn 192.168.56.0/24
```

```
(abc@kali)-[~]
└─$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 01:42 +0000
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.1
Host is up (0.00017s latency).
MAC Address: 0A:00:27:00:00:04 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00021s latency).
MAC Address: 08:00:27:10:6E:11 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up (0.00018s latency).
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.32 seconds
```

Next, perform a basic scan (scanning only the 1,000 most common ports):

```
nmap 192.168.56.102
```

```
(abc@kali)-[~]
└─$ nmap 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 01:45 +0000
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.00013s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds
```

Then, perform a full-port scan (forcing a scan of all ports from 1 to 65535):

```
nmap -p- 192.168.56.102
```

```
(abc@kali)-[~]  
$ nmap -p- 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 01:46 +0000  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.102  
Host is up (0.00011s latency).  
Not shown: 65533 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.96 seconds
```

Scan the services and versions running on Ubuntu (once you obtain the service version numbers, you can search public vulnerability databases for known exploits associated with these versions to launch an attack):

```
nmap -sV 192.168.56.102
```

```
(abc@kali)-[~]  
$ nmap -sV 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 01:50 +0000  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.102  
Host is up (0.00010s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.10 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  http     Apache httpd 2.4.52 ((Ubuntu))  
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 6.47 seconds
```

Perform OS detection (detecting whether the target machine is running Linux):

```
sudo nmap -O 192.168.56.102 (Note: -O uses the uppercase
```

letter 'O', not the number '0')

```
(abc@kali)-[~]
$ sudo nmap -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 01:57 +0000
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.00025s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.54 seconds
```

Vulnerability detection script (using Nmap scripting engine to detect potential risks):

`nmap --script vuln 192.168.56.102`

```
(abc@kali)-[~]
$ nmap --script vuln 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 01:59 +0000
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Stats: 0:00:21 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.51% done; ETC: 02:00 (0:00:00 remaining)
Nmap scan report for 192.168.56.102
Host is up (0.000075s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 22.12 seconds
```

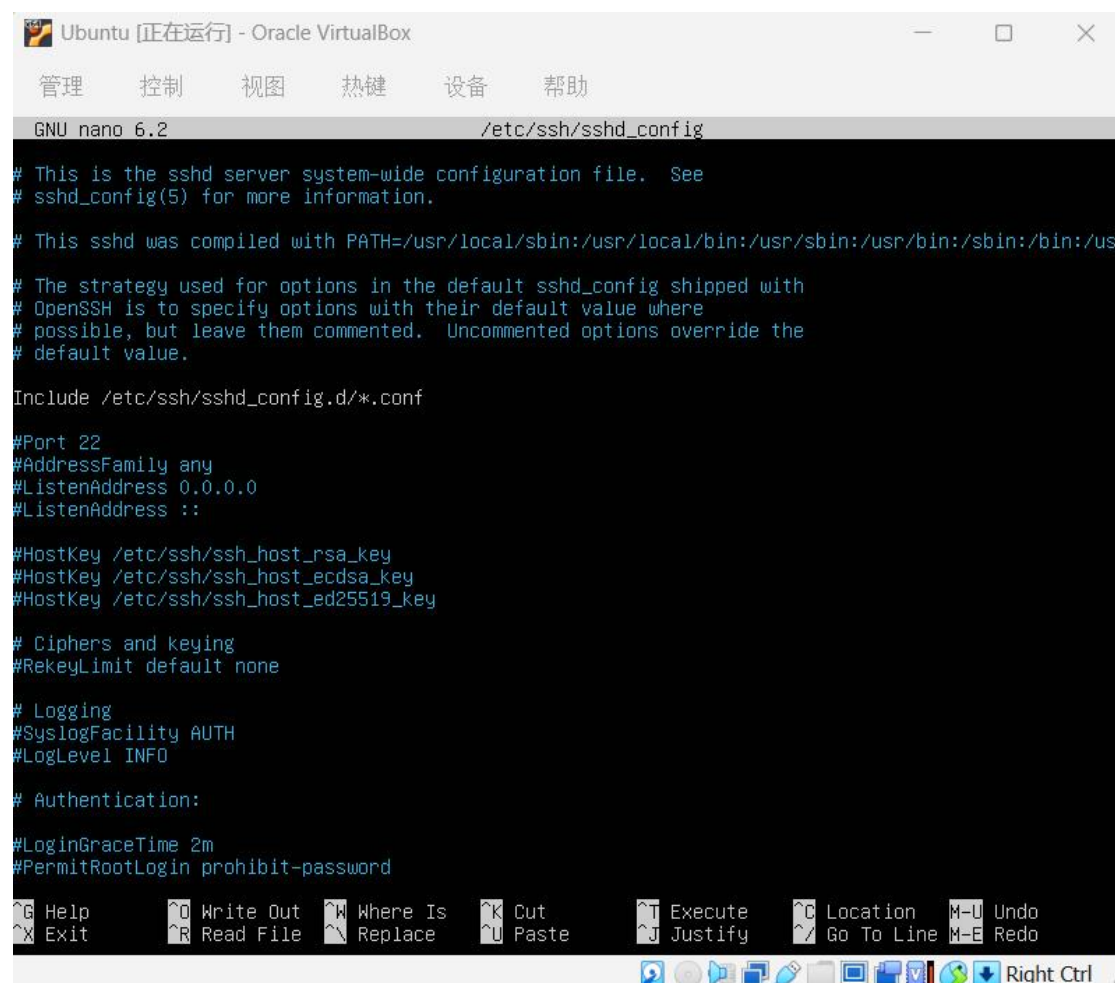
## 5.Security Hardening

Switch back to the Ubuntu virtual machine.

Harden the SSH service (change the default port and disable root login).

Open the SSH configuration file:

```
sudo nano /etc/ssh/sshd_config
```



```
GNU nano 6.2 /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/sbin:/usr/bin

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
```

Modify the following lines directly:

Port 2222

PermitRootLogin no



sudo ufw deny 80/tcp

sudo ufw enable

sudo ufw status

```
abc@ubuntu:~$ sudo ufw allow 2222/tcp
Rules updated
Rules updated (v6)
abc@ubuntu:~$ sudo ufw deny 80/tcp
Rules updated
Rules updated (v6)
abc@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
abc@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
2222/tcp ALLOW Anywhere
80/tcp DENY Anywhere
2222/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) DENY Anywhere (v6)
```

Hide the Apache service version:

Open the Apache security configuration file:

sudo nano /etc/apache2/conf-enabled/security.conf

```
GNU nano 6.2 /etc/apache2/conf-enabled/security.conf
# Debian packages.
#
#<Directory />
#   AllowOverride None
#   Require all denied
#</Directory>

# Changing the following options will not really affect the security of the
# server, but might make attacks slightly more difficult in some cases.
#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
ServerTokens OS
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature On

#
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

Modify the following parameters:

ServerTokens Prod

ServerSignature Off

```
GNU nano 6.2 /etc/apache2/conf-enabled/security.conf *
# Debian packages.
#
#<Directory />
#   AllowOverride None
#   Require all denied
#</Directory>

# Changing the following options will not really affect the security of the
# server, but might make attacks slightly more difficult in some cases.
#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature Off

#
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

Save and exit, then restart Apache:

sudo systemctl restart apache2

## 6.Secondary Scanning

Perform a port scan in the Kali terminal to verify the changes.

```
(abc@kali)-[~]
$ nmap -p- 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 03:13 +0000
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl
ed. Try using --system-dns or specify valid servers with --dns-servers
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Sc
an
SYN Stealth Scan Timing: About 0.72% done
Stats: 0:01:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Sc
an
SYN Stealth Scan Timing: About 7.16% done; ETC: 03:37 (0:22:42 remaining)
```

```
(abc@kali)-[~]
$ nmap -p 22,80,2222,3306 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 03:26 +0000
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.00025s latency).

PORT      STATE      SERVICE
22/tcp    filtered  ssh
80/tcp    filtered  http
2222/tcp  open      EthernetIP-1
3306/tcp  filtered  mysql
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.62 seconds
```

Switch back to Ubuntu to modify the firewall rules and open port 80:

`sudo ufw allow 80/tcp`

```
abc@ubuntu:~$ sudo ufw allow 80/tcp
[sudo] password for abc:
Rule updated
Rule updated (v6)
abc@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
abc@ubuntu:~$
```

Perform a version detection scan in Kali to verify that the version information is hidden and that SSH is running on the new port:

`nmap -sV -p 80,2222 192.168.56.102`

```
(abc@kali)-[~]
$ nmap -sV -p 80,2222 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 05:57 +0000
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.102
Host is up (0.00027s latency).

PORT      STATE      SERVICE      VERSION
80/tcp    open      http         Apache httpd
2222/tcp  open      ssh          OpenSSH 8.9p1 Ubuntu 3ubuntu0.10 (Ubuntu Linux; protocol 2.0)
MAC Address: 08:00:27:D5:8B:1F (Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.38 seconds
```